

Romanization Flattens the Field: Tokenizer-Independent and Downstream Evidence of Script Fragility in Sinhala

Anonymous ACL submission

Abstract

Sinhala is written two ways online. Formal text uses the native script, while everyday messaging is dominated by Romanized Sinhala typed in Latin letters. A recent benchmark reported that language models are roughly 300 times worse at Romanized than at native Sinhala, and that model scale predicts nothing. We show that both conclusions are shaped by the metric. Re-measuring 31 checkpoints with tokenizer-independent bits per byte, the ranking induced by perplexity turns out to be uncorrelated with the ranking induced by bits per byte ($\rho = 0.08$), because perplexity mostly tracks how finely a tokenizer shreds Sinhala. The 300-fold gap becomes 2.14 bits per byte, and scale does predict native-script ability once the confound is removed. We then run the first downstream evaluation of Sinhala script variation, covering three tasks, 9,479 items and ten instruction-tuned checkpoints under both scripts. Romanization compresses a 22.4 point spread in native-script accuracy into 6.7 points, and the loss a model suffers is almost exactly proportional to how much native-script ability it had to begin with ($R^2 = 0.97$). On offensive-language detection, accuracy barely moves while Matthews correlation loses 60% of its value. The effect is not a tokenization-length artifact: Romanized prompts are shorter in tokens for every model we test. Finally, absolute bits per byte predicts downstream script fragility, whereas the bits-per-byte *gap* predicts nothing.

1 Introduction

Sinhala is an Indo-Aryan language with about 17 million speakers and a script of its own (Ranasinghe et al., 2025). Like several South Asian languages, it has an everyday orthographic split. Formal and published writing uses the native Sinhala script, but comments, chat and social media are dominated by Romanized Sinhala, informally called Singlish, in which the same language is typed with Latin letters. The Romanized form

has no standard spelling, so කොහොමද (“how are you”) is typed *kohomada*, *kohomd* or *kohomade* depending on the writer.

This is not a marginal input condition. In a comparable setting for Indian languages, Khullar et al. (2025) found that 83% of Hindi and 73% of Telugu messages sent to a real maternal-health service were Romanized, and that models handled those messages measurably worse. Anyone deploying a Sinhala system for content moderation or user support is therefore serving mostly Romanized input while evaluating on native-script benchmarks.

Rajapakse and Weerasinghe (2026) gave the first systematic measurement for Sinhala. Benchmarking 24 open-weight models with perplexity, they reported a median 312-fold degradation from native to Romanized text and no correlation between parameter count and script handling. Both findings are widely quotable, and both rest on a metric that is not comparable across models. Perplexity is normalised by token count, and token count for a given string depends entirely on the tokenizer (Gao et al., 2020; Magnusson et al., 2024). Sinhala is exactly the setting where that matters, because tokenizers segment it very differently (Petrov et al., 2023; Ahia et al., 2023). The same work also notes that perplexity says nothing about whether a model can still do anything useful, and calls for downstream evaluation.

We take up both problems.

Fixing the metric changes the intrinsic picture.

We recompute bits per byte, per character and per word for 31 checkpoints, 7 of them new relative to prior work, over the same parallel corpus (§4). Our perplexities reproduce the published values, so the differences that follow come from the metric and not the measurement. Across checkpoints, perplexity and bits per byte are uncorrelated ($\rho = 0.08$, $p = 0.68$), and perplexity is instead almost fully explained by tokenizer fertility ($\rho = -0.87$,

$p < 10^{-9}$). The 300-fold gap corresponds to 2.14 bits per byte, a factor of 4.4 per byte. Under bits per byte, scale *does* predict native-script ability ($\rho = -0.55$, $p = 0.001$), though still not Romanized ability.

Romanization flattens rather than merely degrades. Native-script cost varies widely across checkpoints, from 16.3 to 34.1 bits per word, while Romanized cost collapses into a 20.3 to 25.7 band. Twelve of the 31 checkpoints are actually *cheaper* on Romanized text than on native text, simply because they were bad at native Sinhala to start with. A held-out character 4-gram model fit on 400 sentences beats all 31 checkpoints on Romanized Sinhala.

The first downstream evidence for Sinhala.

We evaluate ten instruction-tuned checkpoints on 9,479 items across multiple-choice knowledge, offensive-language detection and physical commonsense, under both scripts, for 189,580 generations (§5). Every comparison is paired and tested with McNemar’s test under Holm correction. The same flattening appears: a 22.4 point spread in native-script accuracy becomes 6.7 points. More precisely, the loss is proportional to how much a model had, $\text{loss} = 0.78 \times \text{headroom above chance}$ with $R^2 = 0.97$, and the same relation holds within the benchmark across 16 domain by difficulty cells. We show the effect is not driven by token-length inflation or by output degeneration.

A usable predictor, and a released transliterator. Absolute native-script bits per byte predicts the downstream gap ($\rho = -0.90$, $p = 0.0009$), while the bits-per-byte *gap*, the quantity prior work reports, predicts nothing ($\rho = -0.07$, $p = 0.86$). Building the Romanized side required a native-to-Latin transliterator, the opposite of the direction most Sinhala tools support. Ours beats three alternatives on all three reference corpora, 755,000 human-romanized items, and we release it as a pip-installable package.

2 Related Work

Sinhala and low-resource evaluation. Sinhala is classified as low-resource and its tooling has grown in a fragmented way (de Silva, 2019; Ranathunga and de Silva, 2022; Joshi et al., 2020). Recent resources are almost entirely native-script: SinhalaMMLU (Pramodya et al., 2025), SOLD (Ranasinghe et al., 2025), the Sinhala split of

Global PIQA (de Silva and Ranathunga, 2026; Chang et al., 2025) and encoder work such as Dhananjaya et al. (2022). The only benchmark treating script as a variable is Rajapakse and Weerasinghe (2026), which we extend.

Romanization gaps elsewhere. Romanized South Asian text has been studied mainly as a resource and identification problem (Roark et al., 2020; Benton et al., 2025). J et al. (2024) show romanization can *help* when models are adapted for it, the mirror image of our zero-shot setting. Closest in spirit is Khullar et al. (2025), who find F1 drops up to 24 points on Romanized clinical queries. We add a paired item-level design with a competence screen, a tokenizer-independent intrinsic account of the same models, and a test of whether the gap is a tokenization artifact.

Why not perplexity. Bits per byte exists so that models with different vocabularies can be compared (Gao et al., 2020), and Paloma adopts it for that reason (Magnusson et al., 2024). Tokenizers are known to treat languages very unequally (Petrov et al., 2023; Ahia et al., 2023). To our knowledge ours is the first work to show that, for a specific language, the substantive conclusions of a published benchmark reverse when the normaliser changes.

3 Experimental Setup

3.1 Script conditions

Every item is evaluated twice, once in the native script as published and once in a Romanized form with identical content, so all comparisons are paired and differ only in orthography. This avoids the confound of using an independently authored Latin-script Sinhala set, of which Global PIQA does publish one.

3.2 Data

Table 1 lists the evaluation sets. The intrinsic corpus is the parallel set released with Rajapakse and Weerasinghe (2026): 500 sentence pairs in which the Romanized side was typed by native speakers rather than generated. This matters, because it means the intrinsic results carry no dependence on our own transliterator. A separate 500-sentence corpus of authentic mixed-script text supports a secondary condition.

For the downstream tasks we use every available item with no sampling. SinhalaMMLU is

Set	Task	Items	Baseline
<i>intrinsic</i>			
Parallel	LM cost, paired	500	—
Mixed	LM cost, single	500	—
<i>downstream</i>			
SinhalaMMLU	4/5-way MCQ	6,879	23.4% chance
SOLD	binary class.	2,500	59.4% majority
Global PIQA	2-way MCQ	100	50.0% chance

Table 1: Evaluation data. Every downstream item is scored under both script conditions, giving 18,958 prompts per checkpoint.

a 6,879-item, curriculum-aligned multiple-choice benchmark in the style of MMLU (Hendrycks et al., 2021), over six domains and three difficulty levels; 2,139 items have five options rather than four, so random guessing scores 23.4% rather than 25%.¹ SOLD supplies the complete 2,500-item test split of a binary offensive-language task where 40.6% of posts are offensive, so always predicting the majority class scores 59.4%. Global PIQA, which follows PIQA (Bisk et al., 2020), contributes the 100 released Sinhala items, 77 of them culturally specific.

3.3 Building the Romanized side

Almost every Sinhala transliteration tool runs Latin to native (De Mel et al., 2025; Sumanathilaka et al., 2025). We need the reverse. We implemented a deterministic grapheme-level transliterator that parses Sinhala as the abugida it is, walking the string and emitting a consonant’s Latin form followed by either an explicit vowel sign, nothing when the virama ේ (U+0DCA) cancels the inherent vowel, or the inherent *a* otherwise. It uses the digraph conventions Sinhala speakers actually type, so *ඪ* is *th* and *ඨ* is *w* rather than *v*. It runs locally with no model or network, covers every input, and is reversible only up to the consonant merges that casual Singlish also collapses, such as *ට/ඪ* both becoming *t/th*. For example *වෙසක් උත්සවයේදී බෞද්ධයින්* becomes *wesak uthsawayeedii baud-dhayin*, and *ප්‍රතිපත්ති*, whose conjunct is joined by a zero-width joiner, becomes *prathipaththi*.

We validated it against 755,000 human-romanized items from the Swa-bhasha Resource Hub (Sumanathilaka et al., 2025): 4,397 authentic social-media sentence pairs, 450,587 words each carrying multiple accepted spellings, and a fixed 300,000-sentence sample of a machine-augmented

¹The public Hugging Face release contains a 1,851-item sample. We obtained the full set from the authors.

Method	CER ↓			chrF Exact		Cov.
	sent.	word	aug.	sent.	word	
Ours	0.182	0.121	0.112	67.8	33.2	100.00
Web tool $v \rightarrow w$	0.182	0.133	0.114	67.8	30.9	99.67
Aksharamukha	0.191	0.148	0.139	63.7	23.9	100.00
Web tool, raw	0.197	0.176	0.153	63.4	20.6	99.67
uroman	0.228	0.228	0.210	55.7	9.7	100.00

Table 2: Transliteration quality against human references (4,397 sentence pairs, 450,587 multi-reference words, 300,000 augmented sentences). Coverage is the share of inputs the method produced any output for.

set used only as a cross-check. Hypotheses are scored against the closest accepted reference, case-folded, using character error rate, chrF (Popović, 2015) and exact match. Table 2 compares our method with Aksharamukha (Rajan), uroman (Hermjakob et al., 2018) and the web romanizer released with de Silva and Ranathunga (2026). Ours has the lowest character error rate on all three corpora, and a paired Wilcoxon test separates it from every alternative at $p < 10^{-90}$ on the two large corpora. The web tool is close once its *v* is rewritten to *w*, which is a spelling convention rather than a transliteration error, but it cannot romanize 17 sequences involving *ඤ* at all and leaves Sinhala characters in 0.8% to 2.5% of its output.

The residual error is one systematic difference rather than phonemic mistakes. All four methods double long vowels far more often than people do, 0.81 per word against 0.12, which is why we produce *bas eka enna thawath welaawak yanawada?* where a human wrote *bus eka enna thawath welawak yanawada?*. Canonicalising that single convention on both sides cuts character error rate to 0.040 on the word corpus. We keep the uncanonicalised form, since over-regular spelling is conservative: it is closer to a standard than real Singlish, so it should be easier for a model.

3.4 Models

The intrinsic evaluation covers 31 decoder-only checkpoints from 350M to 14.7B parameters, spanning the Llama (Grattafiori et al., 2024), Qwen (Yang et al., 2024, 2025), Gemma (Gemma Team et al., 2024), Mistral, BLOOM (BigScience Workshop et al., 2023), OPT, Pythia (Biderman et al., 2023) and Phi (Abdin et al., 2024) families. Twenty-four match Rajapakse and Weerasinghe (2026), and we add three newer base checkpoints

and four instruction-tuned counterparts. Appendix Table 7 gives every identifier.

The downstream evaluation needs models that follow a response format, so it uses the ten instruction-tuned checkpoints in the pool, from 1.1B to 9B parameters. Two of them turn out to be unable to follow the format at all in the native-script condition, which we handle explicitly in §3.7 rather than by dropping them.

3.5 Prompting and scoring

All downstream prompts are zero-shot and greedily decoded with a 40-token limit. We chose one template from three candidates using a pilot of 3,240 generations over nine checkpoints, then applied it to every dataset and both script conditions. What a paired comparison needs is not an optimal template but one that does not favour a script, and in the pilot the winning template was the same under both scripts for all three datasets. Answers are extracted by pattern match, and anything unparseable is scored as wrong and reported separately as an invalid rate (Appendix A).

3.6 Statistics

Both conditions score the same items, so we use McNemar’s exact or continuity-corrected test (McNemar, 1947; Dror et al., 2018) and report the discordant pair counts. Within each dataset we control the family-wise error rate over checkpoints with Holm’s procedure (Holm, 1979). Confidence intervals on gaps, macro-F1 differences and Matthews correlation differences come from 10,000 paired item bootstrap resamples. For SOLD we lead with Matthews correlation (Matthews, 1975; Chicco and Jurman, 2020) because the classes are imbalanced and accuracy is close to the majority baseline. Item-level effects are also fitted as logistic regressions with checkpoint fixed effects and cluster-robust standard errors by item.

3.7 The competence screen

A model at chance in the native-script condition cannot lose anything by being given Romanized input, so a small gap for such a model is evidence of nothing. Reporting it alongside real effects would understate the phenomenon. We therefore mark a checkpoint as *competent* on a task when its native-script score is above the guessing baseline at $p < 0.01$ under a Poisson-binomial null, and when it

does not put more than 90% of its answers on a single option. Five of ten checkpoints pass on Sinhala-MMLU, five have a non-trivial native-script signal on SOLD, and none pass on Global PIQA. All ten appear in every table.

4 The Intrinsic Picture

4.1 Metrics

For a text s we accumulate the model’s total negative log-likelihood $\text{NLL}(s)$ in nats and divide by a unit of the text itself rather than by its tokenization:

$$\text{BPB} = \frac{\sum_s \text{NLL}(s)}{\ln 2 \sum_s |s|_{\text{bytes}}}, \quad (1)$$

and likewise per character (BPC) and per whitespace word (BPW). We pool numerator and denominator over the corpus rather than averaging per-sentence ratios, so short sentences do not dominate. Bytes are the standard choice (Gao et al., 2020), but BPW is sharpest for cross-script comparison, because a sentence and its romanization contain the same number of words while Sinhala UTF-8 spends 2.65 bytes per character against 1.00 for Latin.

As a sanity check, our recomputed perplexities reproduce the published values for all 24 shared checkpoints, with median absolute deviation 0.05% on native and 0.004% on Romanized text. Everything below therefore reflects the metric, not a different run.

4.2 Perplexity ranks models by their tokenizers

Figure 1 makes the central methodological point. Across the 31 checkpoints, native-script perplexity and native-script bits per byte are statistically unrelated ($\rho = 0.08$, $p = 0.68$; Romanized $\rho = 0.13$, $p = 0.48$). Only one of the five best checkpoints by perplexity is also among the five best by bits per byte. Gemma-2-9B ranks 26th of 31 on perplexity and 2nd on bits per byte, and Qwen3.5-9B-Base moves from 28th to 6th.

The cause is visible in panel (b). Perplexity is almost fully explained by how many tokens a tokenizer spends on a Sinhala word ($\rho = -0.87$, $p < 10^{-9}$), which ranges from 4.3 for the Qwen3.5 and Gemma tokenizers to 14.2 for models that fall back to bytes. Spreading the same loss over three times as many tokens lowers per-token perplexity without the model knowing anything more. So a

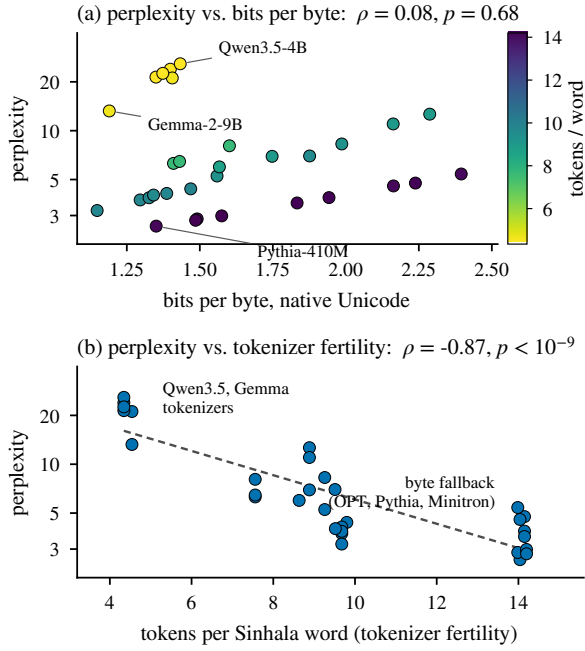


Figure 1: Perplexity does not measure what it appears to. (a) Across 31 checkpoints, native-script perplexity is unrelated to native-script bits per byte. (b) It is instead almost fully determined by how many tokens the tokenizer spends per Sinhala word.

headline perplexity comparison across model families on Sinhala is largely a comparison of tokenizers, and reported perplexity ratios inherit the same problem.

Two conclusions follow. First, the reported 312-fold degradation is mostly a tokenization effect. Our median ratio is 294-fold, but in bits per byte the same gap is 2.14 bits, a factor of 4.4 per byte, and in bits per word it is 1.11 bits. Second, the finding that scale does not predict script handling does not survive. Parameter count is uncorrelated with native perplexity ($\rho = 0.15$, $p = 0.42$) but clearly correlated with native bits per byte ($\rho = -0.55$, $p = 0.001$) and with mixed-script bits per byte ($\rho = -0.57$, $p = 0.0008$). Scale does buy native-script ability. What it does not buy is Romanized ability, where the correlation stays insignificant ($\rho = -0.28$, $p = 0.13$).

4.3 Romanization flattens the field

Figure 2(a) shows the effect that motivates the rest of the paper. Native-script cost spans 16.3 to 34.1 bits per word across checkpoints, a spread of 17.7 bits. Romanized cost spans 20.3 to 25.7, a spread of 5.4. Regressing one on the other, a full bit per word of native-script improvement buys only 0.14 bits on Romanized text (95% CI [0.05, 0.25], $R^2 =$

Model	Native Unicode		Romanized	
	BPB	# LMs <	BPB	# LMs <
character 1-gram	1.815	22	3.977	24
character 2-gram	1.352	7	3.063	0
character 3-gram	1.162	1	2.685	0
character 4-gram	1.269	2	2.636	0
character 5-gram	1.474	15	2.947	0
character 6-gram	1.680	21	3.381	3
best checkpoint	1.148	–	3.290	–
median checkpoint	1.486	–	3.805	–

Table 3: Held-out character n -gram models against the 31 checkpoints. “# LMs <” counts checkpoints that achieve lower bits per byte than the n -gram model. No checkpoint beats a character bigram on Romanized Sinhala.

0.19). Twelve of the 31 checkpoints are cheaper on Romanized than on native Sinhala, and which ones is entirely predictable: the better a checkpoint is natively, the more it loses ($\rho = -0.88$ between native cost and the change, $p < 10^{-9}$). Romanized Sinhala is a regime in which model quality has largely stopped mattering.

The mixed-script condition behaves quite differently and reproduces the earlier finding. Native and mixed bits per byte are strongly rank-correlated ($\rho = 0.94$) with a median gap of only 0.49 bits, against $\rho = 0.41$ for Romanized. Interleaving native tokens with Latin ones is a mild perturbation. Replacing the script entirely is not.

4.4 How much is that in absolute terms?

Relative statements leave open whether models are merely worse or genuinely uninformed. As a reference we fit add- α character n -gram models on the same corpus under five-fold cross-validation, so each score is out of sample, and convert to bits per byte. This is a cheap in-domain model and the comparison favours it in that respect, but the size of the difference is still informative. Table 3 reports it, and Figure 4 in the appendix plots it. On Romanized Sinhala the best n -gram model reaches 2.64 bits per byte and even a character bigram reaches 3.06, while the best of 31 checkpoints reaches 3.29 and the median 3.81. Not one checkpoint beats the bigram. On native Sinhala the picture is ordinary, with the character trigram at 1.16 and one checkpoint below it. Whatever these models know about Sinhala, they know it about the script rather than the language.

5 Downstream Consequences

Table 4 reports every checkpoint on every task.

5.1 Multiple-choice knowledge

Four of the five competent checkpoints lose significantly under Romanization after Holm correction. Qwen3.5-9B falls from 44.1% to 28.7%, a 15.5 point drop (95% CI [14.0, 16.9], 1,892 against 828 discordant pairs), Qwen3.5-4B loses 10.6 points, Llama-3.1-8B-Instruct 5.4 and Qwen2-7B-Instruct 2.0. The fifth, Hormoz-8B, starts only 2.5 points above chance and loses 0.8 points, which is not significant. Pooled over the cohort the gap is 6.9 points (95% CI [6.3, 7.5], $n = 34,395$ paired items), and a logistic fit with checkpoint fixed effects and item-clustered errors puts the odds ratio for Romanized input at 0.68 ($p < 10^{-30}$).

Read as levels the result is starker. Figure 2(b) shows native-script accuracy spanning 22.4 points while Romanized accuracy spans 6.7. Romanized accuracy never exceeds 28.7% against a 23.4% chance baseline, so the best checkpoint keeps 5.2 of its 20.7 points of headroom, and regressing Romanized on native accuracy gives a slope of 0.22 (95% CI [0.12, 0.54]). The six checkpoints showing no gap are the six already at chance.

5.2 You can only lose what you have

The relationship in the previous paragraph is remarkably tight. Writing headroom as native accuracy minus chance, a linear fit across the nine parseable checkpoints gives

$$\text{loss} = 0.78 \times \text{headroom} - 0.87, \quad (2)$$

with $R^2 = 0.97$ and $p = 1.9 \times 10^{-6}$ (Figure 3a). Romanization removes roughly four fifths of whatever native-script ability a checkpoint had.

A single regression on nine points is thin evidence, so we test the same relation inside the benchmark, where the units are items rather than models. Splitting SinhalaMMLU into its 16 non-empty domain by difficulty cells and computing headroom and loss per cell for the competent cohort gives a slope of 0.58 with $R^2 = 0.83$ and $p = 9.8 \times 10^{-7}$ (Spearman 0.85). The two decompositions are independent, one across models and one across content, and they agree.

Table 5 shows the strata. Loss falls monotonically with difficulty, 8.6 points on Easy items, 7.2 on Medium and 5.2 on Hard, and the interaction with difficulty is significant in the item-clustered

logistic fit ($p = 0.007$). Hard items lose less because models are nearer chance on them in both conditions. By domain, Social Science loses most at 11.8 points and is also where native-script accuracy is highest, while Humanities, the largest domain at 3,341 items, loses 4.8 and has lower native accuracy. The pattern is uncomfortable for benchmark design: the harder and less well-served a slice of a benchmark is, the smaller the script effect it can reveal.

5.3 Offensive-language detection: accuracy hides the damage

SOLD is where the choice of metric decides the conclusion. Accuracy falls by only 1.6 to 11.3 points against a 59.4% majority baseline, so three checkpoints look almost unaffected. Matthews correlation says otherwise (Figure 3b). Qwen3.5-9B goes from 0.280 to 0.069, keeping 24% of its signal, Qwen2-7B-Instruct from 0.148 to 0.042, keeping 29%, and Llama-3.1-8B-Instruct from 0.259 to 0.105, keeping 40%. Four of the five checkpoints with a real native-script signal lose significantly under Holm correction and the median keeps 40%. The mechanism is a retreat toward trivial behaviour: Qwen2-7B-Instruct more than doubles its rate of predicting the offensive class, 10.5% to 23.4%, without becoming more accurate, and Hormoz-8B does the same from 9.8% to 20.7%. A deployment tracking accuracy here would see a 5 point wobble while over half the usable signal disappeared.

5.4 Global PIQA is underpowered, and says so

No checkpoint is above chance on the native-script condition of Global PIQA, so the screen admits none, and with 100 items and a median of 36 discordant pairs no comparison is significant. The direction is consistent, seven of nine parseable checkpoints dropping, but a sign test gives $p = 0.18$ and pooling the cohort gives a 4.6 point gap at $p = 0.10$. We report this as a null result rather than corroboration. A 100-item set cannot resolve effects of the size we measure elsewhere, which is worth saying plainly given how often small multilingual splits are read as evidence.

5.5 Two things it is not

Not token fragmentation. The natural explanation for a Romanized penalty is that romanization shatters words into more tokens. For Sinhala the

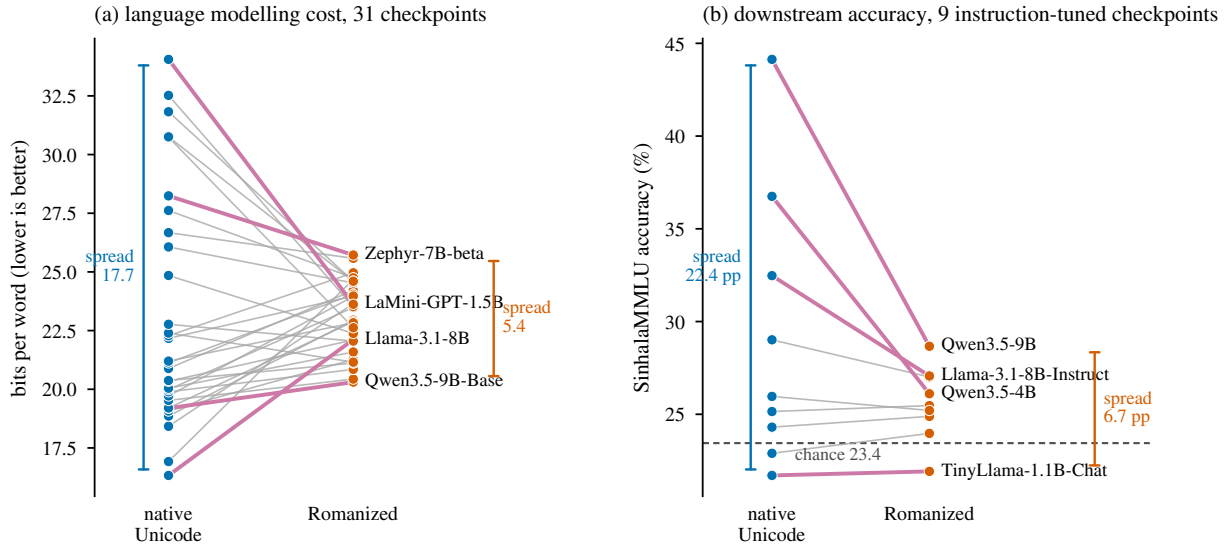


Figure 2: Romanization flattens the field. (a) Language modelling cost in bits per word for 31 checkpoints. Word counts are identical within a pair, so the two columns are directly comparable. Native-script cost varies by 17.7 bits across checkpoints; Romanized cost varies by 5.4, and crossing lines mark the twelve checkpoints that are cheaper on Romanized text. (b) SinhalaMMLU accuracy for the instruction-tuned checkpoints. A 22.4 point spread becomes 6.7, and no checkpoint finishes more than 5.3 points above chance. LaMini-GPT-1.5B is omitted from (b) because it produces unparseable output on 95.9% of native-script items; it appears in Table 4.

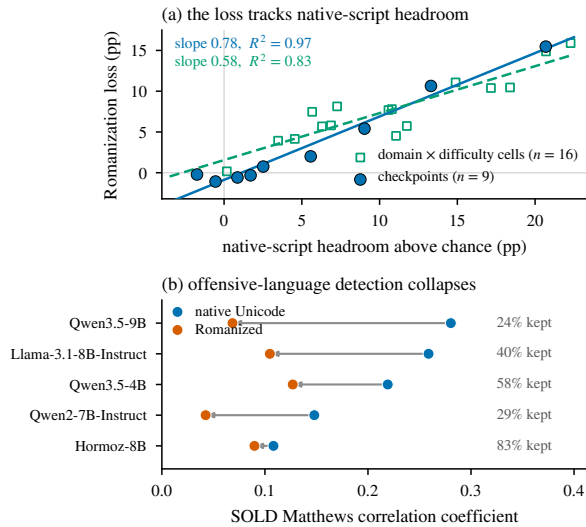


Figure 3: (a) Romanization loss against native-script headroom, across checkpoints and independently across the 16 domain by difficulty cells of SinhalaMMLU. (b) Matthews correlation on SOLD for the five checkpoints with a real native-script signal, with the percentage retained under Romanization.

opposite holds. The native script costs a median 9.5 tokens per word against 2.09 for its romanization, because these tokenizers largely fall back to bytes on Sinhala. Every one of the ten checkpoints receives *fewer* tokens in the Romanized condition on all three tasks, a median ratio of 0.48 on SinhalaMMLU (column τ of Table 4). Models get shorter,

more compactly encoded input and do worse. Splitting items into terciles by how much romanization inflates tokens gives no monotone pattern either, with losses of 5.9, 7.5 and 7.2 points. The gap reflects what is missing from pretraining, not how the input is chopped up. BLOOM illustrates this: its pretraining corpus covers 46 languages and Sinhala is not one of them (Laurençon et al., 2023; Big-Science Workshop et al., 2023), and it sits at the bottom on native Sinhala while being unremarkable on Romanized text.

Not output degeneration. A model could score at chance by emitting one constant answer. For the competent cohort the share of answers on the most frequent option is 37.0% natively and 38.6% under romanization (Wilcoxon $p = 0.63$), and among the five competent checkpoints the invalid rate never exceeds 0.03% in either condition. The models keep answering across the option set. They are simply wrong.

6 Do Intrinsic Metrics Predict Downstream Fragility?

Can a cheap intrinsic measurement stand in for an expensive downstream one? Table 6 answers this for the nine checkpoints measured both ways. Absolute native-script bits per byte predicts the SinhalaMMLU gap ($\rho = -0.90$, $p = 0.0009$) and

Checkpoint	SinhalaMMLU (6,879 items)						SOLD (2,500 items)				Invalid %			
	Acc. %					p_{Holm}	macro-F1		MCC		p_{Holm}			
	B	U	R	Δ	95% CI		U	R	U	R		U	R	τ
TinyLlama-1.1B-Chat	1.1	21.7	21.9	-0.2	[-0.5,0.1]	.8273	6.4	24.2	-0.019	-0.043	1.0000	1.5	1.0	0.45
LaMini-GPT-1.5B	1.5	0.9	18.8	-17.9	[-18.8,-16.9]	<.0001	5.1	28.3	0.000	0.024	1.0000	95.9	16.4	0.31
StableLM-Zephyr-3B	2.8	22.9	24.0	-1.1	[-2.3,0.1]	.4275	38.6	38.9	0.036	0.001	.9288	3.8	2.6	0.32
SmolLM3-3B	3.1	25.1	25.5	-0.3	[-1.0,0.4]	.8273	29.7	29.9	0.025	0.017	1.0000	0.0	0.0	0.48
Qwen3.5-4B†	4.0	36.7	26.1	+10.6	[9.3,12.0]	<.0001	60.2	55.1	0.219	0.127	.0014	0.0	0.0	0.72
Zephyr-7B-beta	7.2	24.3	24.9	-0.6	[-1.7,0.5]	.8273	39.7	39.6	0.060	0.060	1.0000	2.1	0.6	0.49
Qwen2-7B-Instruct†	7.6	29.0	27.0	+2.0	[0.8,3.2]	.0047	49.9	50.3	0.148	0.042	<.0001	0.0	0.0	0.51
Llama-3.1-8B-Instruct†	8.0	32.5	27.1	+5.4	[4.1,6.7]	<.0001	60.1	52.6	0.259	0.105	<.0001	0.0	0.0	0.45
Hormoz-8B†	8.0	26.0	25.2	+0.8	[-0.5,2.0]	.8273	47.9	51.9	0.108	0.090	1.0000	0.0	0.0	0.50
Qwen3.5-9B†	9.0	44.1	28.7	+15.5	[14.0,16.9]	<.0001	63.1	52.0	0.280	0.069	<.0001	0.0	0.0	0.72

Table 4: Downstream results, ordered by size. U is native Unicode, R is Romanized, Δ is the paired difference in points and p_{Holm} is McNemar’s test corrected across checkpoints within a task. MCC is Matthews correlation and its p_{Holm} comes from a paired bootstrap. τ is the ratio of Romanized to native input tokens on SinhalaMMLU, below one for every checkpoint. Invalid % is the share of unparseable generations, on SinhalaMMLU. † marks checkpoints passing the competence screen of §3.7. Global PIQA is in Appendix Table 8.

Stratum	Items	U %	R %	Δ	p_{Holm}
<i>by difficulty</i>					
Easy	1,851	37.7	29.1	8.6	<.0001
Medium	2,503	35.9	28.7	7.2	<.0001
Hard	2,525	28.5	23.2	5.2	<.0001
<i>by domain</i>					
Social Science	1,059	38.5	26.7	11.8	<.0001
Business Studies	463	35.2	27.1	8.0	<.0001
Stem	614	34.2	26.2	8.0	<.0001
Other	1,014	32.5	24.5	8.0	<.0001
Humanities	3,341	32.7	27.9	4.8	<.0001
Language	388	29.4	24.7	4.7	.0002

Table 5: SinhalaMMLU by stratum, pooled over the five competent checkpoints. Item counts are per checkpoint. Domains are ordered by loss.

the loss of Matthews correlation on SOLD ($\rho = -0.82$, $p = 0.007$). Lower bits per byte means a larger script gap, which is the floor effect of §5.2 seen from the intrinsic side.

The negative result is the more useful one. The bits-per-byte *gap* between conditions, the natural analogue of the degradation ratio prior work reports, predicts nothing ($\rho = -0.07$, $p = 0.86$), and neither does the perplexity ratio ($\rho = -0.53$, $p = 0.14$). The widely reported quantity is the wrong one to act on. To anticipate how much a model will suffer on Romanized Sinhala, measure how good it is at native Sinhala, tokenizer-independently.

7 Discussion

Three things follow for practice. First, a benchmark reporting perplexity across model families on a script that tokenizers handle unevenly is re-

Intrinsic predictor	MMLU gap		SOLD Δ_{MCC}	
	ρ	p	ρ	p
Unicode BPB (level)	-0.90	.0009	-0.82	.0072
Romanized BPB (level)	-0.85	.0037	-0.92	.0005
BPB gap (Rom. – Uni.)	-0.07	.8647	-0.40	.2861
Unicode BPW (level)	-0.90	.0009	-0.82	.0072
perplexity ratio	-0.53	.1392	-0.42	.2646
parameter count	+0.65	.0567	+0.49	.1768

Table 6: Spearman correlations between intrinsic measurements and downstream script effects, over the nine parseable checkpoints. Levels predict; gaps and ratios do not.

porting tokenizer behaviour, and for Sinhala this inverts both the model ranking and the conclusion about scale. Bits per byte costs nothing extra on the same forward passes. Second, script robustness cannot be read off a group average: six of our ten checkpoints sit at chance and contribute a zero gap by construction, and the same trap operates inside a benchmark, where the hardest strata reveal the smallest effects. Any robustness claim needs a competence screen and a stated guessing baseline. Third, the models best at native Sinhala have the most to lose, so picking by native-script scores and serving Romanized users selects for the largest drop.

The flattening result also says where effort should go. Romanized ability is nearly constant across 31 checkpoints spanning a 42-fold parameter range, and none beats a character bigram fit on 400 sentences. That is missing data, not something scale will fix: the remedies are transliteration before inference and Romanized pretraining data.

Limitations

One language. Everything here is Sinhala. The mechanism we describe, a floor effect that makes robustness gaps proportional to native ability, should generalise to any language with a dominant Romanized register, but we have not shown that. Hindi, Urdu, Nepali, Tamil and Arabic dialects are the obvious tests, and [Khullar et al. \(2025\)](#) report compatible magnitudes for the first three.

The Romanized side of the downstream sets is synthetic. We transliterate native-script items rather than collecting human-typed Romanized versions of them, because no such parallel resource exists at this scale. Our transliterator is the best of four on 755,000 human references, and its residual error is a spelling regularity that should make the text easier rather than harder, but it is still not what a person would type. The intrinsic results do not share this limitation, since their Romanized side was typed by native speakers, and they show the same flattening. A human-written Romanized version of even a few hundred SinhalaMMLU items would settle the question.

Open weights only, and small ones. We evaluate open checkpoints up to 9B for the downstream tasks. Frontier proprietary models are much stronger on Sinhala ([Pramodya et al., 2025](#)) and therefore have far more headroom, so Equation 2 predicts larger absolute losses for them. That prediction is untested here.

Zero-shot only. We use one template, zero-shot, greedy. This measures what a user gets out of the box, not the ceiling. Few-shot prompting, transliterate-then-infer, or Romanized fine-tuning could all narrow the gap, and we do not measure by how much.

Two checkpoints cannot do the task. LaMini-GPT-1.5B and TinyLlama-1.1B-Chat produce unparseable output on most native-script items of at least one task. Their apparent improvement under Romanization is an artifact of that failure, not a script effect, which is why the competence screen exists.

Ethics Statement

All datasets are public research releases used within their licences and for evaluation only, which matters for Global PIQA, whose licence forbids training. SOLD contains offensive social media

text; it is already pseudonymised upstream, with user mentions replaced by placeholders, and we neither collected nor re-annotated any of it. We add no new human annotation and no new data collection.

The practical risk our results speak to is deployment. A moderation or support system validated on native-script Sinhala and serving Romanized users will look far better in evaluation than in the field, and on SOLD that failure is invisible in accuracy while more than half the usable signal is gone. Publishing the size of that gap, and a screen that stops averaging from hiding it, is the useful response. Our transliterator collapses several Sinhala consonant distinctions, matching casual Singlish practice, so it is a tool for building evaluation data and not for rendering names or other text where those distinctions carry meaning.

Acknowledgments

Omitted for anonymous review.

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A Prompt Template

One template is used for every dataset and both script conditions. Multiple choice, with the option count n and the domain name inserted where available:

```
This is a multiple-choice question related to the {domain}. Choose the correct or most appropriate answer from answers 1/.../n for the following question. Do not repeat the question or text in your response. Respond with only the number (1/.../n) and nothing else.
Question: {question}
1. {option 1} ... n. {option n}
Answer:
```

Binary classification:

```
Classify the following Sinhala text as either 'NOT' (not offensive) or 'OFF' (offensive). Do not repeat the question or text in your response. Respond with only 'NOT' or 'OFF' and nothing else.
Text: {text}
Answer:
```

Prompts are rendered through each checkpoint’s own chat template. For checkpoints that emit reasoning traces by default we disable that behaviour identically in both script conditions. Decoding is greedy with at most 40 new tokens.

B Checkpoints

Table 7 lists all 31 checkpoints with their Hugging Face identifiers and their tokenizer fertility on the parallel corpus. Note the consistency of Romanized fertility, between 1.9 and 2.4 tokens per word, against a range of 4.3 to 14.2 on the native script.

C Global PIQA and the n -gram Reference

Table 8 gives the per-checkpoint Global PIQA results omitted from Table 4, and Figure 4 plots the n -gram comparison of Table 3.

Checkpoint	B params	Hugging Face identifier	tokens / word	
			Unicode	Romanized
OPT-350M	0.35	facebook/opt-350m	14.1	2.4
Pythia-410M	0.41	EleutherAI/pythia-410m	14.0	2.2
BLOOM-560M	0.56	bigscience/bloom-560m	8.9	1.9
BLOOM-1B1	1.10	bigscience/bloom-1b1	8.9	1.9
TinyLlama-1.1B-Chat‡	1.10	TinyLlama/TinyLlama-1.1B-Chat-v1.0	9.8	2.4
Llama-3.2-1B	1.20	meta-llama/Llama-3.2-1B	9.7	2.2
Cerebras-GPT-1.3B	1.30	cerebras/Cerebras-GPT-1.3B	14.0	2.3
OPT-1.3B	1.30	facebook/opt-1.3b	14.1	2.4
LaMini-GPT-1.5B‡	1.50	MBZUAI/LaMini-GPT-1.5B	14.0	2.3
Qwen1.5-1.8B	1.80	Qwen/Qwen1.5-1.8B	7.6	2.1
OPT-2.7B	2.70	facebook/opt-2.7b	14.1	2.4
StableLM-Zephyr-3B‡	2.80	stabilityai/stablelm-zephyr-3b	14.0	2.2
BLOOM-3B	3.00	bigscience/bloom-3b	8.9	1.9
SmolLM3-3B‡	3.10	HuggingFaceTB/SmolLM3-3B	9.5	2.1
Llama-3.2-3B	3.20	meta-llama/Llama-3.2-3B	9.7	2.2
Qwen3.5-4B-Base	4.00	Qwen/Qwen3.5-4B-Base	4.3	2.0
Qwen3.5-4B‡	4.00	Qwen/Qwen3.5-4B	4.3	2.0
Mistral-7B-v0.3	7.20	mistralai/Mistral-7B-v0.3	9.3	2.4
Zephyr-7B-beta‡	7.20	HuggingFaceH4/zephyr-7b-beta	9.3	2.4
Qwen2-7B	7.60	Qwen/Qwen2-7B	7.6	2.1
Qwen2-7B-Instruct‡	7.60	Qwen/Qwen2-7B-Instruct	7.6	2.1
Llama-3.1-8B	8.00	meta-llama/Meta-Llama-3.1-8B	9.7	2.2
Llama-3.1-8B-Instruct‡	8.00	meta-llama/Llama-3.1-8B-Instruct	9.7	2.2
Hormoz-8B‡	8.00	mann-e/Hormoz-8B	8.6	2.1
Minitron-8B-Base	8.40	nvidia/Minitron-8B-Base	14.2	1.9
Gemma-7B	8.50	google/gemma-7b	4.5	2.0
Qwen3.5-9B-Base	9.00	Qwen/Qwen3.5-9B-Base	4.3	2.0
Qwen3.5-9B‡	9.00	Qwen/Qwen3.5-9B	4.3	2.0
Gemma-2-9B	9.20	google/gemma-2-9b	4.5	2.0
Mistral-Nemo-Base-2407	12.20	mistralai/Mistral-Nemo-Base-2407	14.2	2.2
Phi-4-14B	14.70	microsoft/phi-4	9.5	2.1

Table 7: All evaluated checkpoints, with Hugging Face identifiers and tokenizer fertility on the parallel corpus. ‡ marks the ten used for the downstream evaluation. Native-script fertility varies by more than a factor of three across tokenizers, whereas Romanized fertility is nearly constant.

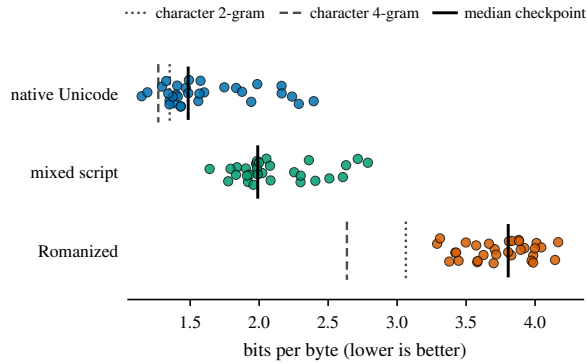


Figure 4: Bits per byte for all 31 checkpoints in each script condition, with held-out character n -gram references. The mixed-script corpus is not parallel, so no reference is shown for it.

D Full Intrinsic Results

Table 9 gives every metric for every checkpoint in all three script conditions, corpus-pooled as in Equation 1.

E Reproducibility

Every number in this paper is produced by scripts that read only the per-item model outputs, so the tables and figures cannot drift from the data. We release the frozen evaluation sets with both script conditions paired per record, all per-item generations for every checkpoint, the analysis and figure scripts, and the transliterator: <https://anonymous.4open.science/r/sinhala-script-robustness>.

The transliterator is also available as `pip install sinlipi`, documented at <https://pypi.org/project/sinlipi>.

Intrinsic evaluation scores 46,500 sequences (31 checkpoints, 1,500 sequences each). Downstream evaluation produces 189,580 generations (10 checkpoints, 9,479 items, two script conditions). Both use fp16 on commodity dual-T4 GPUs. Greedy decoding and a fixed template make the downstream runs deterministic up to GPU non-determinism, and the transliteration and data

Checkpoint	Acc. %		disc.		p_{Holm}	Inv. %	
	U	R	Δ	b/c		U	R
TinyLlama-1.1B-Chat	40	33	+7	12/5	1.0000	18	28
LaMini-GPT-1.5B	0	36	-36	0/36	<.0001	100	28
StableLM-Zephyr-3B	45	44	+1	15/14	1.0000	0	0
SmolLM3-3B	53	51	+2	14/12	1.0000	0	0
Qwen3.5-4B	57	49	+8	22/14	1.0000	0	0
Zephyr-7B-beta	48	50	-2	18/20	1.0000	0	0
Qwen2-7B-Instruct	45	42	+3	21/18	1.0000	0	0
Llama-3.1-8B-Instruct	53	48	+5	22/17	1.0000	0	0
Hormoz-8B	47	48	-1	13/14	1.0000	0	0
Qwen3.5-9B	53	45	+8	25/17	1.0000	0	0

Table 8: Global PIQA, 100 items. “disc. b/c” are the discordant pair counts entering McNemar’s test. No checkpoint is above chance in the native-script condition and no comparison survives correction, so we treat this task as underpowered rather than as corroboration.

preparation steps are fully deterministic with byte-identical outputs recorded as SHA-256 digests.

Checkpoint	B	Native Unicode			Romanized			Mixed script		
		PPL	BPB	BPW	PPL	BPB	BPW	PPL	BPB	BPW
Llama-3.1-8B	8.0	3.22	1.148	16.33	970.5	3.584	22.12	6.36	1.641	19.66
Gemma-2-9B	9.2	13.23	1.190	16.92	4726.1	3.999	24.68	48.99	1.792	21.47
Llama-3.2-3B	3.2	3.74	1.296	18.42	1560.7	3.831	23.65	7.40	1.776	21.27
Llama-3.1-8B-Instruct‡	8.0	3.86	1.326	18.86	1255.5	3.718	22.95	7.94	1.839	22.03
Phi-4-14B	14.7	4.01	1.341	19.07	1149.1	3.446	21.27	8.08	1.832	21.94
Qwen3.5-9B-Base	9.0	21.39	1.350	19.19	1250.1	3.290	20.30	78.56	1.958	23.46
Pythia-410M	0.4	2.58	1.351	19.20	1245.0	3.707	22.88	4.91	1.978	23.69
Qwen3.5-9B‡	9.0	22.56	1.373	19.52	1307.8	3.311	20.43	84.24	1.990	23.83
Llama-3.2-1B	1.2	4.11	1.387	19.72	1718.6	3.881	23.96	8.54	1.903	22.79
Qwen3.5-4B-Base	4.0	23.92	1.399	19.89	1512.7	3.378	20.85	83.32	1.985	23.77
Gemma-7B	8.5	21.13	1.406	19.99	3138.9	3.805	23.48	66.00	1.930	23.11
Qwen2-7B	7.6	6.29	1.410	20.05	1357.0	3.498	21.59	14.11	1.919	22.98
Qwen2-7B-Instruct‡	7.6	6.46	1.431	20.34	1611.2	3.581	22.10	15.45	1.985	23.77
Qwen3.5-4B‡	4.0	25.84	1.433	20.37	1663.1	3.422	21.12	90.71	2.023	24.23
TinyLlama-1.1B-Chat‡	1.1	4.38	1.469	20.89	1014.2	3.923	24.21	8.92	1.980	23.71
Mistral-Nemo-Base-2407	12.2	2.80	1.486	21.12	1938.4	3.896	24.04	4.63	1.912	22.90
Cerebras-GPT-1.3B	1.3	2.86	1.491	21.20	1040.2	3.700	22.83	5.23	2.052	24.58
Mistral-7B-v0.3	7.2	5.26	1.559	22.17	921.4	3.884	23.97	10.15	1.984	23.76
Hormoz-8B‡	8.0	5.98	1.567	22.28	4175.8	4.045	24.96	13.58	2.083	24.94
Minitron-8B-Base	8.4	2.98	1.575	22.39	2118.6	3.428	21.15	5.08	2.000	23.95
Qwen1.5-1.8B	1.8	8.07	1.601	22.77	1584.9	3.573	22.05	17.60	2.079	24.90
BLOOM-3B	3.0	6.95	1.748	24.85	3258.5	3.626	22.38	15.90	2.254	26.99
OPT-2.7B	2.7	3.59	1.833	26.06	1052.8	3.974	24.53	6.26	2.300	27.54
SmolLM3-3B‡	3.1	6.98	1.876	26.67	5373.0	4.145	25.58	13.89	2.301	27.56
OPT-1.3B	1.3	3.87	1.942	27.62	1121.9	4.010	24.75	6.82	2.407	28.83
Zephyr-7B-beta‡	7.2	8.28	1.986	28.24	1515.6	4.167	25.72	15.76	2.360	28.27
BLOOM-1B1	1.1	11.00	2.162	30.74	3558.5	3.666	22.62	24.52	2.607	31.22
StableLM-Zephyr-3B‡	2.8	4.57	2.163	30.76	2129.2	3.986	24.60	7.51	2.509	30.05
OPT-350M	0.3	4.75	2.238	31.82	1075.9	3.986	24.60	8.15	2.630	31.50
BLOOM-560M	0.6	12.64	2.287	32.52	4915.0	3.810	23.52	28.01	2.715	32.52
LaMini-GPT-1.5B‡	1.5	5.41	2.395	34.06	1322.0	3.827	23.62	9.46	2.788	33.39

Table 9: Corpus-pooled intrinsic metrics for all 31 checkpoints, sorted by native-script bits per byte. ‡ marks the checkpoints also evaluated downstream. Bits per character are omitted because they are a fixed multiple of bits per byte within a script, $2.651\times$ for native Sinhala and $1.003\times$ for its romanization. Note that the PPL and BPB orderings disagree almost completely.